Relja Brdar

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Education

Faculty of Technical Sciences, University of Novi Sad *Oct 2023 – Jun 2027**
BSc in Software Engineering

Gymnasium "Jovan Jovanović Zmaj", Novi Sad *Sep 2019 – Jun 2023*
Special Course for Students Gifted in Mathematics

Petnica Science Center
Alumni, Mathematics Seminar

Work Experience

Data Scientist @ Ominimo *Sep 2026 – Present*

- Work across data engineering and data science: preparing datasets, improving and optimizing data pipelines for insurance pricing and market analysis.
- Train pricing models and run market analyses using **gradient boosting**, **SVMs**, and **clustering** algorithms.

SWE Internship @ SuperPlane *Mar 2026 – Aug 2026*

***SuperPlane**: open-source, AI-first control plane for production infrastructure — $2.6M pre-seed (Credo Ventures)*

- Contributed across the workflow-engine platform: integrations, core components, devops, and observability, agentic features, evaluation and optimisation including a **10x reduction** in LLM token use for the in-app agent.
- Bootstrapped **Runner**, a private core component of the workflow engine, with a teammate: built the broker → fleet-manager → runner (VM) architecture to execute **shell**, **Python**, and **JS** jobs on cloud VMs, covering reconciliation logic, WebSocket task handling/routing, CI/CD, dynamic machine allocation, pool sizing, OTel metrics, and rate limiting.

SWE Internship @ NimbusTech *Sep 2025 – Nov 2025*

- Full-stack intern building distributed apps with **Go**, **Python**, and **React**. Delivering features across backend, frontend, and CI/CD, integrating cloud infrastructure on **AWS**, databases, and LLM-powered services.

Projects

Tileflow [link] *Python, C++*

End-to-end deep learning compiler and runtime built from scratch featuring a Python-based IR with optimization passes, a static scheduler with double-buffering memory management, and a pipelined C++ execution engine with lock-free thread coordination and hand-written AVX2 SIMD kernel.

Distributed NoSQL Database Engine [link] *Go*

Originated as coursework for "Advanced Algorithms and Data Structures"

NoSQL database engine in Go (LSM-Tree, WAL for ACID compliance, Bloom filters), extended into a distributed, leaderless system with pipelined, truly concurrent read/write paths, Merkle-tree based antientropy, and quorum-based replication with SWIM gossip membership.

Hackathons

- **First place** at the regional Frontier AI Hackathon organized by Garaža, sponsored by **Perplexity**, **Tenderly**, **Solflare**, **Kortix** and others. *(Dec 2025)*
- **Ominimo** insurance hackathon finalist - tackled a real-world insuretech pricing problem with gradient-based ML; received a data scientist offer from the first Serbian unicorn. *(Mar 2026)*
- Competed at HackUPC in Barcelona, the biggest European student hackathon (700+ competitors). *(Apr 2026)*

Algorithms & Data Structures

Over 350 problems solved on leetcode.

Languages

English (C2, Cambridge Advanced Exam - A Grade) | **Serbian** (Native)